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Is there a risk-return trade-off in cryptocurrency markets? The case of Bitcoin



Walid M.A. Ahmed

Department of Business Administration, Faculty of Business, Ain Shams University, P.O. Box 11566, Cairo, Egypt

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ABSTRACT

This study examines the link between returns and volatility of Bitcoin, at both contemporaneous and intertemporal levels, employing high-frequency data. The intraday price variability is proxied by four different measures, namely realized variance, jump variation, downside realized semivariance, and negative signed jump variation. The empirical analysis suggests that all realized volatility proxies have a significant and negative contemporaneous relation with Bitcoin returns. On the other hand, there is weak evidence of a negative intertemporal relation between returns and realized variance, jump variation, and downside realized semivariance. Accordingly, the existence of a positive risk-return trade-off in Bitcoin markets seems to be unsubstantiated. The findings are robust, even after controlling for a number of relevant determinants of the price formation process of Bitcoin.

1. Introduction

The empirical nature of the relationship between risk and returns continues to be one of the most intriguing, yet controversial, research topics in financial economics and related fields, due to its vital implications for asset-pricing analysis, portfolio selection, market efficiency, capital budgeting decisions, and corporate risk management practices. Under a set of confining assumptions, standard theory predicts a positive proportionality of the market risk premium, anticipated by risk-averse investors, to the conditional variance of the market. The static (unconditional) capital asset pricing model (CAPM), developed by Sharpe (1964) and Lintner (1965), implies that expected returns on a risky asset are only driven by the covariance of that asset's returns with those of the benchmark market portfolio. Thus, within the one-period CAPM world, the expected returns of an asset are contemporaneously positively correlated with its market beta risk. Nonetheless, such a static single-factor model framework fails to consider conditioning factors, which may cause an asset's beta to be of a time-varying behavior. As a viable extension of the static CAPM to a multi-period context, Merton's (1973) two-factor intertemporal CAPM states that the equilibrium expected rate of return of a given security is positively proportional to both its covariance with the market portfolio and its covariances with each of the economy's state variables proxying for changes in the investment opportunity set. Several studies (e.g., Guo and Whitelaw, 2006; Scruggs, 1998; Turtle, Buse, & Korkie, 1994) underscore the importance that asset pricing models account for a hedge component of expected returns to ensure a positive risk-return trade-off. On the other hand, Backus and Gregory (1993) argue that the relation between risk premiums and conditional second moments of asset returns can be increasing, decreasing, flat, or even nonmonotonic, depending on both the preferences of the representative agent and the stochastic structure of the economy. Similarly, Rossi and Timmermann (2010) indicate that simple linear models are unlikely to be adequate to capture the association between expected returns and conditional volatility. Hence, the theoretical risk premium-conditional variance relation may not be described as a positive linear function at all

E-mail address: Walid.Ahmed@abmmc.edu.qa.

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times.

On the empirical front, there is an ever-expanding body of literature that endeavors to detect the presence, or lack thereof, of a positive trade-off between risk and returns over time and across markets, utilizing variants of alternative asset-pricing models and different estimation techniques. However, evidence on the sign of the coefficient of relative risk aversion seems to be far from being consistent. At one extreme, substantial research documents a significant positive risk-return relation (e.g., [Bali & Peng, 2006](#); [Bollerslev, Osterrieder, Sizova, & Tauchen, 2013](#); [Chiang, Li, & Zheng, 2015](#); [Ghysels, Santa-Clara, & Valkanov, 2005](#); [Lanne & Saikkonen, 2007](#); [Müller, Durand, & Maller, 2011](#)). At the other extreme, many studies establish a significant negative relation (e.g., [Ang, Hodrick, Xing, & Zhang, 2009](#); [Badshah, Frijns, Knif, & Tourani-Rad, 2016](#); [Brandt & Kang, 2004](#); [Jin, 2017](#); [Lettau & Ludvigson, 2010](#)). Other works find a statistically weak or insignificant association (e.g., [Baillie & DeGennaro, 1990](#); [Campbell & Hentschel, 1992](#); [French, Schwert, & Stambaugh, 1987](#)). Moreover, there is a raft of literature that considers a time-varying framework to uncover the risk-return relation. For instance, [Ghysels, Guérin, and Marcellino \(2014\)](#) examine the state-dependent relation between returns and conditional volatility, using a Markov-switching specification. The results show a negative (positive) linkage in the first (second) regime, thus confirming the instability of the risk-return relation across time. [Wu and Lee \(2015\)](#) demonstrate that the nature of the intertemporal risk-return relation differs between bull and bear cycles of the US markets. That is, the association is statistically significant and positive (negative) during bull (bear) market conditions.

Unquestionably, the low-volatility anomaly (i.e., low-volatility assets generate greater risk-adjusted returns than do their high-volatility counterparts) runs counter to the primary intuition that high-volatility assets must deliver commensurately high returns. [Baker, Bradley, and Wurgler \(2011\)](#) regard the long-term outperformance of low-volatility and low-beta stock portfolios as perhaps the most leading anomaly in finance. Consequently, a stream of research attempts to propose candidate explanations, in order to resolve such a provocative paradox. Notable examples embrace the leverage aversion theory ([Asness, Frazzini, & Pedersen, 2012](#); [Black, 1972](#)), the money-illusion hypothesis ([Cohen, Polk, & Vuolteenaho, 2005](#); [Modigliani & Cohn, 1979](#)), the aggregate disagreement theory ([Hong & Sraer, 2016](#)), the sentiment-induced mispricing hypothesis ([Shen, Yu, & Zhao, 2017](#)), and the reference-dependent preference theory ([Wang, Yan, & Yu, 2017](#)). In practice, since the risk-return trade-off may not always hold in real-world markets, some portfolio managers undertake the so-called betting-against-beta/volatility strategies designed to profit from such a mispricing of risk.¹

Surprisingly, despite its practical connection with many investment decisions, the risk-return relation has not hitherto been analyzed in the context of the burgeoning virtual currency markets. Over the recent years, cryptocurrencies, such as Bitcoin, Ethereum, Ripple and Libra, have rapidly developed from being merely the preserve of a niche group of enthusiasts to becoming a popular point of interest for both academic and professional communities across the globe. Virtual currencies are primarily based upon an amalgam of cryptographic principles and peer-to-peer protocols that guarantee safe and fault-tolerant transactions. The emergence of cryptocurrencies, as an extremely secure payment mechanism for electronic transactions and as an investment asset class, has absorbed the attention of e-commerce managers, investors, regulators, and policy makers, given their far-reaching effects. A *de facto* standard of cryptocurrencies, Bitcoin seems to be the world's first digital currency and blockchain database, which came into existence in 2009 as the brainchild of an anonymous programmer, or possibly a group of programmers, using the alias Satoshi Nakamoto. Aside from being the most prominent since its market debut, Bitcoin has steadily positioned itself as the largest of all cryptocurrencies, both in terms of volume traded and market capitalization. According to [blockchain.com](#), a chief global software platform for digital assets, the trading volume of Bitcoin increased exponentially from only 637.00 US dollars up to 2.66 billion US dollars, and the total market value of Bitcoin supply in circulation multiplied more than 127 thousand times, from 2.35 million US dollars up to 297.39 billion US dollars, over the period January 2011–December 2018. This meteoric growth in Bitcoin value and trading volume is most probably fueled by its innovative features. Interestingly, in late 2017, two of the world's largest futures exchanges, Chicago Mercantile Exchange (CME) and Chicago Options Exchange (CBOE), were granted approval from the US Commodity Futures Trading Commission (CFTC) to list Bitcoin futures contracts on their respective trading platforms. Earning the cryptocurrency ecosystem at large more liquidity and legitimacy, this milestone move enables Bitcoin to compete on an equal footing with major physical commodities, such as precious metals, crude oil, and agricultural goods.

Global popularity and adoption notwithstanding, Bitcoin is notorious for its excessive volatility, so much so that many mainstream investors are apparently shy of entering the Bitcoin exchange market. In fact, the lack of appropriate legal and regulatory frameworks for decentralized virtual currencies has put Bitcoin prices at the mercy of speculative frenzies. For instance, Bitcoin markets have been on a roller coaster ride since the end of 2017. According to [blockchain.com](#), Bitcoin shed almost 84 % of its value, plummeting from an all-time high of nearly 20,000 US dollars on 17 December 2017 to as low as 3193 US dollars on 16 December 2018, causing consternation among traders. As noted by [Tu and Meredith \(2015\)](#), there are some instances in which Bitcoin loses roughly a third or even half of its value in the space of a trading day. [Blau \(2017\)](#) finds that the estimated volatility of Bitcoin is nearly twice as large as that of 51 different sovereign currencies, over the period July 2010–June 2014. [Alvarez-Ramirez, Rodriguez, and Ibarra-Valdez \(2018\)](#) show that Bitcoin markets are not uniformly efficient, given the existence of price bubbles over various time scales. [Aysan, Demir, Gozgor, and Lau \(2019\)](#) establish that changes in the global geopolitical risks index are positively and negatively correlated with the price volatility and returns of Bitcoin, respectively. [Yermack \(2015\)](#) argues that Bitcoin fails to fulfill the traditional functions of fiat currencies, viz., a medium of exchange, a store of value, and a unit of account. [Shahzad, Bouri, Roubaud,](#)

¹ [Frazzini and Pedersen \(2014\)](#) coin the term “betting-against-beta” (BAB) to describe a profitable strategy that exploits the low risk anomaly by purchasing leveraged low-beta (i.e., safe) stocks and holding a short position in deleveraged high-beta (i.e., risky) stocks. The authors show that the BAB strategy generates significant positive risk-adjusted returns.

Kristoufek, and Lucey (2019) find that Bitcoin acts as a weak safe-haven for Chinese stock markets. Uyar and Kahraman (2019) find that Bitcoin/US dollar exchange rates are about 6 times riskier than the exchange rates of Swiss franc/US dollar, Sterling pound/US dollar, Japanese yen/US dollar, Canadian dollar/US dollar, Australian dollar/US dollar, and New Zealand dollar/US dollar.

On reflection, such eccentric properties could possibly render the risk-return profile of Bitcoin dissimilar from those of other traditional financial assets, thus warranting more intensive and rigorous investigation. The main aim of this study is to provide empirical evidence on the relation between Bitcoin returns and volatility, at both contemporaneous and intertemporal levels, utilizing high-frequency price data. More explicitly, the central questions that this work seeks to investigate are:

- i What does the contemporaneous risk-return relation of Bitcoin exchange rates look like?
- ii What does the intertemporal risk-return relation of Bitcoin exchange rates look like?
- iii Does the relation, if any, vary with different measures of risk?
- iv Are common global economic factors relevant sources of variation in Bitcoin exchange rates?

By addressing the above issues, the study makes three chief contributions to the emerging economic literature on Bitcoin. First, due to the nascent nature of the cryptocurrency phenomenon, relatively less attention has been devoted to exploring the price dynamics of Bitcoin. Furthermore, to date, no studies have specifically examined the trade-off between risk and returns in the Bitcoin market, a lacuna in the literature that this study attempts to fill. To the author's best knowledge, Bouri, Azzi, and Dyhrberg (2017) seems to be only published work that analyzes the return-volatility relation for Bitcoin. However, a major methodological difference between the two studies is that Bouri et al. adopt an asymmetric-GARCH framework, taking no notice of the global factors that could possibly affect the price dynamics of Bitcoin, whereas the current study estimates the risk-return relation in the context of the Generalized Method of Moments (GMM), after controlling for a set of influential factors common to the world financial system. Moreover, although GARCH-type models can aptly explain volatility clustering, they are not suited to account for sharp sudden jumps in asset returns. Second, the vast majority of research on the risk-return association (e.g., Arshanapalli, Fabozzi, & Nelson, 2013; Baillie & DeGennaro, 1990; Bouri, Azzi et al., 2017; Brandt & Kang, 2004; Ghysels et al., 2005; Jiang & Lee, 2014; Ludvigson & Ng, 2007; Wang & Yang, 2013) tends to measure volatility utilizing low-frequency price data (daily, weekly or even monthly), which may overlook important intraday information and critical aspects of the price process. This work improves on these studies by relying on a high-frequency intraday dataset, which is very likely to provide a more accurate picture of volatility. Third, the intraday price variability of Bitcoin in this study is proxied by four alternative measures (i.e., realized variance, jump variation, downside realized semivariance, and negative signed jump variation), in order to assess the sensitivity of the risk-return relation to different volatility measures. The empirical findings could prove useful for traders, investors, regulators, and other market participants, when it comes to understanding the dynamics and peculiarities of Bitcoin exchange rates.

The remainder of this study is structured as follows. Section 2 sheds light on the nature of Bitcoin. Section 3 outlines the econometric framework. Data description and preliminary analysis are given in Section 4, whilst the empirical findings are presented in Section 5. The penultimate section investigates the robustness of the results, followed by a summary and conclusions in the final section.

2. The nature of Bitcoin

What does Bitcoin represent? Is it a form of currency like the US dollar, an investment asset like stocks, or a safe-haven commodity like gold?. These questions have recently sparked the interest of academics, practitioners, and policymakers, since their answers can quite well serve to broaden our overall appreciation of the distinct features and specificities of Bitcoin and cryptocurrencies at large. Relevant literature is by no means unanimous about the genuine nature of virtual currencies, though. In essence, as shown by Nakamoto (2008), Bitcoin was invented as an electronic payment system whose integrity is assured by a cryptographic hash-based proof-of-work algorithm, rather than a trusted central authority. The original goal of cryptocurrencies is to provide an anonymous, seamless, and incorruptible form of transactions amongst the parties involved. Unlike traditional fiat money, Bitcoin is a completely decentralized currency independent of any national government or central banking authorities. Furthermore, due to the fact that the maximum number of Bitcoin units is capped to only 21 million, there is not a limitless supply of Bitcoin, which implies that it has a deflationary nature (Adhami, Giudici, & Martinazzi, 2018; Baur, Hong, & Lee, 2018).

A typical currency performs at least three basic functions: (i) a medium of exchange; (ii) a unit of account; and (iii) a store of value. Concerning the first function, thanks to the advantages that Bitcoin has over traditional fiat-based payment systems (e.g., reduced transaction fees, independence from central banks, user anonymity, absence of third-party intermediaries), an increasing multitude of consumers, merchants, and institutions are willing to adopt it as an acceptable method of payment for goods and services. However, more realistically, there are a number of challenges that militate against endorsing Bitcoin as a universal medium of exchange. For instance, the global commercial usage of Bitcoin and its variants combined is undeniably dwarfed by that of traditional forms of payment. Alfieri, Burlacu, and Enjolras (2019) and Turpin (2014) emphasize that the lack of legal status and clear-cut regulations hinders Bitcoin from widespread acceptance. Another obstacle is that the sole channel available to procure bitcoins is cryptocurrency exchanges, which necessitate that users be able to undertake Bitcoin mining activities (Yermack, 2015). With regard to the second function, Bitcoin's usefulness as a unit of account or numéraire turns out to be impeded by a variety of real-life barriers. For example, prices of most convenience goods are disconcertingly minuscule when expressed in the equivalent value of Bitcoin (e.g., a chocolate bar may cost about 0.00012 in Bitcoin terms). Vendors, who accept Bitcoin as a valid form of payment, have to frequently change the prices of items offered for sale, because of the wild fluctuations of Bitcoin exchange rates. Additionally, the

discrepancy in Bitcoin prices across various exchanges leaves consumers too confused about the right price (Turpin, 2014, Yermack, 2015). Lastly, with respect to the third function, some scholars tend to be skeptical of Bitcoin's viability as a store of value, due to its extreme volatility and its potential cybersecurity vulnerabilities. Cheah and Fry (2015) document the presence of significant speculative bubbles in Bitcoin prices. They also find that the long-term fundamental value of Bitcoin is zero. Baur, Hong et al. (2018) indicate that Bitcoin might be utilized as a substitute to official currencies experiencing rapid depreciation, thus acting as a temporary store of value. Nelson (2018) emphasizes that digital currency-related technologies are more susceptible to fraud and cyber risks than are conventional fiat-based payment systems.

On the other hand, some market analysts and scholars (e.g., Popper, 2015) view Bitcoin as digital gold, since Bitcoin and gold have similar characteristics. For instance, a common aspect of Bitcoin and gold is non-centrality, which means that there exists no central authority in charge of controlling their electronic mining and physical extraction, respectively. In some mature economies (e.g., the US, Canada, Finland), Bitcoin is categorized as a commodity on a par with gold. Both have no identifiable cash flows, but can be used as a currency hedge (e.g., Urquhart & Zhang, 2019; Zagaglia & Marzo, 2013), a portfolio diversifier (e.g., Hillier, Draper, & Faff, 2006; Platanakis & Urquhart, 2019), and a safe-haven investment in turbulent times (e.g., Ciner, Gurdgiev, & Lucey, 2013; Feng, Wang, & Zhang, 2018). As far as differences are concerned, gold has a physical embodiment; its overall supply is unpredictable and has no date limit, while Bitcoin is merely a digital representation of a virtual currency with no physical presence; its overall supply is predictable and engineered to max out in year 2040 (Baur, Hong et al., 2018). In this respect, Selgin (2015) suggests that Bitcoin is a synthetic commodity money, because it is as scarce as commodity money (e.g., gold), but, analogous to fiat money (e.g., the US dollar), it has no fundamental value.

A further strand of literature indicates that Bitcoin shares considerable similarities with financial investment assets. For example, the empirical investigation into Bitcoin price and trading dynamics reveals price clustering (e.g., Hu, McInish, Miller, & Zeng, 2019; Urquhart, 2017), pricing inefficiency (e.g., Nadarajah & Chu, 2017; Sensoy, 2019), long memory of returns (e.g., Al-Yahyaee, Mensi, & Yoon, 2018; Bariviera, 2017), tail risk behavior (Osterrieder & Lorenz, 2017; Phillip, Chan, & Peiris, 2018), and relatively lower bid-ask spreads (e.g., Dyhrberg, Foley, & Svec, 2018; Kim, 2017). Feng et al. (2018) points out that if Bitcoin is considered as an alternative asset class, one must examine its risk-return profile and its ability to act as a hedge or a diversifier for other assets. From a portfolio diversification perspective, Bitcoin appears to be an attractive investment option, due to its weak correlations with traditional financial assets, precious metals, and energy commodities (e.g., Bouri, Molnár, Azzi, Roubaud, & Hagfors, 2017; Guesmi, Saadi, Abid, & Ftiti, 2019; Platanakis & Urquhart, 2019). On the other side, there is consistent evidence that Bitcoin prices seem to have a non-trivial speculative bubble component. Baek and Elbeck (2015) document that Bitcoin prices are about 26-fold more volatile than equity prices, bringing up concerns that Bitcoin serves much more like a speculative asset than a proper currency. The results of Bouoiyour and Selmi (2015) and Ciaian, Rajcaniova, and Kancs (2016)) show that investor speculations are a key determinant of Bitcoin price formation. Baur, Hong et al. (2018) find that Bitcoin is held, for the most part, as a speculative investment, rather than as an alternative currency or a medium of exchange. Corbet, Lucey, and Yarovaya (2018)) also report similar results.

3. Methodology

3.1. Volatility measures

Over the years, modeling and forecasting the variability of asset prices has been the subject of rigorous research in academia and industry, due to its important implications for asset valuation, portfolio construction, and risk management decisions. As argued by Poon and Granger (2003), volatility forecasts can serve as an early warning signal for the vulnerability of financial markets and the economy at large. Given the inherent latent feature of volatility, a proper modeling approach based upon strong parametric assumptions is necessary to provide a measure of the true volatility process. Indeed, before intraday high-frequency data become conveniently accessible, there have been an immense variety of traditional methods, including, for instance, Threshold Autoregressive (TAR) models, Exponentially Weighted Moving Average (EWMA) method, GARCH-class models, and Stochastic Volatility (SV) models, that count on historical close-to-close returns to formulate volatility forecasts. Notwithstanding their merits and popularity, these daily- and lower-frequency return-based models fail to adequately capture the full range of information embedded in asset price movements. As prices generally tend to exhibit alternating episodes of high and low fluctuations of differing length and intensity during a trading day, it is of great importance to take such intraday price uncertainty into account when characterizing the volatility behavior of asset returns. Horst, Rodriguez, Gzyl, and Molina (2012) point out that intraday price swings can be quite large, even though closing prices could be exactly the same as, or not substantially far from, their opening counterparts during a trading day. Therefore, the informational content of intraday prices appears to convey a more realistic picture of volatility. Thanks to the availability of high-frequency transaction data across an ever-expanding spectrum of financial assets in recent years, a raft of research has employed intraday return data to construct *ex post* volatility measures.

To start with, consider that Bitcoin price dynamics belongs to a continuous-time jump diffusion process that exhibits both stochastic volatility and occasional discontinuous jumps as follows:

$$p_t = p_0 + \int_{t=1}^t \alpha_s ds + \int_{t=1}^t \sigma_s dW_s + \sum_{k=1}^{q_t} J_k \quad (1)$$

Eq. (1) can equivalently be expressed as an Itô stochastic differential equation as follows:

$$dp_t = \alpha_t dt + \sigma_t dW_t + J_t dq_t, \quad 0 \leq t \leq T \quad (2)$$

where p_t is the logarithmic price of Bitcoin at time t , α_t is a continuous drift of finite variation, σ_t is a non-negative stochastic volatility process, and W_t denotes a standard Brownian motion. $J_t dq_t$ is a pure jump part, where q_t refers to the jump-counting process normalized such that $dq_t = 1$ indicates a jump at time t with potentially time-varying intensity λ_t and $dq_t = 0$ otherwise, and J_t represents the magnitude of the corresponding jump in Bitcoin's return process at time t if $dq_t = 1$. Of note, for infinitesimal intervals, the drift is negligible and therefore the martingale component is generally the dominant contributor to the return variation (Chan, Gray, & Campen, 2008). In this regard, modeling Bitcoin prices with both a continuous (persistent) component and a discontinuous jump component means that the total return volatility can likewise be separated into a diffusion volatility process, σ_t , and a jump volatility process, J_t . The former represents the smooth variation of the price, whereas the latter captures large price discontinuities resulting from unanticipated information releases.

Nevertheless, as underscored by Andersen, Bollerslev, and Diebold (2007), it is econometrically problematic to differentiate between the key components of the price variation in the jump-diffusion model. Alternatively, the theory of quadratic variation (QV , hereafter) enables this separation to be made in a non-parametric framework. Specifically, the QV of Eq. (2) is formally defined as:

$$QV_t = \int_{t-1}^t \sigma_s^2 ds + \sum_{k=1}^{q_t} J_{t,k}^2 = IV_t + \sum_{k=1}^{q_t} J_{t,k}^2 \quad (3)$$

Thus, Eq. (3) attributes the total variation in Bitcoin's return process to the diffusive volatility, which is also known as the integrated volatility (IV_t) over a given time interval ($t-1, t$), and the aggregation of squared jumps. In reality, however, due to the lack of a continuum of price observations, empirical research uses prices recorded at discrete time points, which implies that QV is not directly observable. In a seminal work, Andersen and Bollerslev (1998) propose a model-free volatility measure, the realized variance (RV , hereafter), which, in its standard set-up, is built on the sum of high-frequency squared returns over a discrete time interval of typically one day. To determine intraday returns within the active part of a trading day t , suppose a sample of Bitcoin prices over a period of T days, where $t \in \{1, 2, \dots, T\}$, and prices are observed M times per day at equally spaced intervals of length $\Delta t = t_i - t_{i-1}$, $1 \leq i \leq M$. If $P_{t,i}$ represents the log-price level at time t_i , then the corresponding return, $R_{t,i}$, is given as follows:

$$R_{t,i} = P_{t,i} - P_{t,i-1}, \quad i = 1, 2, \dots, M, \quad t = 1, 2, \dots, T \quad (4)$$

Accordingly, the RV estimator is specified as:

$$RV_t = \sum_{i=1}^M R_{t,i}^2 \quad (5)$$

This non-parametric measure is the sample analogue of. As established by Comte and Renault (1998) and Barndorff-Nielsen and Shephard (2002), among others, RV converges uniformly in probability to total QV as the time space between observations goes to zero (i.e., $\Delta t \rightarrow 0$) or, equivalently, the sampling frequency goes to infinity (i.e., $M \rightarrow \infty$)

$$\lim_{M \rightarrow \infty} RV_t \equiv QV_t = \int_{t-1}^t \sigma_s^2 ds + \sum_{k=1}^{q_t} J_{t,k}^2 \quad (6)$$

To distinguish between the individual contributions of non-jump and jump parts, Barndorff-Nielsen and Shephard (2004) propose the realized bipower variation (RBV , hereafter) measure, which is defined as:

$$RBV_t = \mu_1^{-2} \frac{M}{M-1} \sum_{i=2}^M |R_{t,i-1}| |R_{t,i}| \quad (7)$$

where $\mu_1 = E(|Z|) \equiv \sqrt{2/\pi} \approx 0.79788$ and Z is a standard normally distributed random variable. The term $M/(M-1)$ denotes a finite sample correction factor. Under certain assumptions, Barndorff-Nielsen, Shephard, and Matthias (2006) and Jacod (2008) show that, as $M \rightarrow \infty$, RBV is a consistent estimator of the IV_t of the continuous sample path component, even in the presence of jumps

$$\lim_{M \rightarrow \infty} RBV_t \equiv IV_t = \int_{t-1}^t \sigma_s^2 ds \quad (8)$$

Thus, the difference between RV and the jump-robust RBV can serve to identify the jumps contribution to the total return variation process. Hence, from Eqs. (6) and (8), the jump variation component (JV , hereafter) can be estimated as follows:

$$\lim_{M \rightarrow \infty} (RV_t - RBV_t) \equiv \sum_{k=1}^{q_t} J_{t,k}^2 = JV_t \quad (9)$$

Following Andersen et al. (2007) and Tauchen and Zhou (2011), the present study adopts the relative measure of JV , which is defined as

$$JV_t = (RV_t - RBV_t)/RV_t \quad (10)$$

Next, to test for significant jumps, the empirical analysis employs the jump detection test statistic (Z), described in Huang and Tauchen (2005) and Tauchen and Zhou (2011).

Although Eq. (9) helps to identify the jump portion, JV , it tells nothing about the potential asymmetric behavior of that portion. In practice, asset prices may show much more sensitivity to downward than upward market movements. Investors tend to be even more vigilant over downside losses than upside gains, and, during bearish market conditions, they demand a risk premium for holding assets that are highly covariant with the market. Based upon the concept of RV , [Barndorff-Nielsen, Kinnebrock, and Shephard \(2010\)](#) introduce measures of downside risk (i.e., negative realized semivariance) and upside risk (i.e., positive realized semivariance). Downside and upside realized semivariance (i.e., bad and good volatilities) for day t are given by:

$$RS_t^- = \sum_{i=1}^M R_{t,i}^2 I(R_{t,i} < 0) \xrightarrow{P} 0.5 \int_{t-1}^t \sigma_s^2 ds + \sum_{k=1}^{q_t} J_{t,k}^2 I(J_{t,k} < 0) \quad (11)$$

$$RS_t^+ = \sum_{i=1}^M R_{t,i}^2 I(R_{t,i} > 0) \xrightarrow{P} 0.5 \int_{t-1}^t \sigma_s^2 ds + \sum_{k=1}^{q_t} J_{t,k}^2 I(J_{t,k} > 0) \quad (12)$$

where $I(\cdot)$ is an indicator function that takes the value of unity if the respective arguments ($R_{t,i} < 0$) and ($R_{t,i} > 0$) are true, and zero otherwise. RS^- (RS^+) is a measure of downside (upside) risk that captures the variation resulting solely from negative (positive) intraday returns. [Nolte and Xu \(2015\)](#) indicate that if asset returns follow a pure diffusion process without jumps, either signed type of RS is expected to converge to half the IV . Nonetheless, if asset returns are driven by a jump diffusion process, the RS^- and RS^+ may not be of equal size. [Patton and Sheppard \(2015\)](#) show that the use of indicator functions in Eqs. (11) and (12) makes it possible to extract the signed jump variation (SJV , hereafter) as follows:

$$SJV_t \equiv RS_t^+ - RS_t^- = \sum_{k=1}^{q_t} J_{t,k}^2 I(J_{t,k} > 0) - \sum_{k=1}^{q_t} J_{t,k}^2 I(J_{t,k} < 0) \quad (13)$$

Additionally, to cast more light on the roles of positive and negative jumps in the volatility process, [Wen et al. \(2016\)](#) propose to decompose the daily signed jump variation, SJV_t , into negative, SJV_t^- , and positive, SJV_t^+ , parts as follows:

$$SJV_t^- = (RS_t^+ - RS_t^-) I\{(RS_t^+ - RS_t^-) < 0\} \quad (14)$$

$$SJV_t^+ = (RS_t^+ - RS_t^-) I\{(RS_t^+ - RS_t^-) > 0\} \quad (15)$$

Using 5-minute price data for three equity market indexes and a set of 105 individual stocks, [Patton and Sheppard \(2015\)](#) find that the information content of negative realized semivariance is considerably larger than that of positive realized semivariance. The results also indicate that the effect of a price jump on volatility rests upon the sign of the jump, with negative (positive) jumps leading to higher (lower) future volatility. [Wen et al. \(2016\)](#) demonstrate that realized volatility, continuous sample path variation, negative realized semivariance, signed jump and signed semi-jumps offer more information for forecasting the volatility of crude oil futures than do discontinuous jump variation and positive realized semivariance. Using a GARCH-M model with autoregressive conditional jump intensity, [Arshanapalli et al. \(2013\)](#) document that the jump component plays a significant role in explaining the relation between equity returns and risk, through its influence on the conditional variance. [Bali, Demirtas, and Levy \(2009\)](#) report evidence of a positive and significant association between lagged downside risk and the portfolio returns on NYSE/AMEX/Nasdaq stocks. [Sévi \(2013\)](#) finds that both realized variance and realized semivariance estimators are positively related to future stock returns. Based on panel data from 27 emerging markets, [Atilgan and Demirtas \(2013\)](#) document a positive cross-sectional relationship between one-month-ahead expected returns and downside risk.

3.2. Econometric framework

This study aims to assess the contemporaneous and intertemporal relations between risk and returns in Bitcoin markets. To this end, the analysis draws on four different proxies of intraday price variability, namely RV , JV , RS^- , and SJV^- , as defined in Eqs. (5), (10), (11), and (14), respectively. The purpose of this exercise is to verify whether the nature of each relation is sensitive to the various definitions and components of volatility.

The baseline time-series regression exploring the contemporaneous association takes the following form:

$$R_t = \alpha + \sum_{i=1}^q \xi_i R_{t-i} + \beta V_t + \varepsilon_t, \quad \varepsilon_t \sim (0, \sigma_\varepsilon^2) \quad (16)$$

where R_t is the Bitcoin's daily rate of return, α is an intercept term, V_t is a scalar predictor denoting a specific measure of realized volatility, β is a single slope coefficient portraying the linear relation between each measure of volatility and returns of Bitcoin, and ε_t is a stochastic disturbance term assumed to be independently and identically distributed with zero mean and unit variance. A statistically significant and positive value of β suggests that higher volatility induces higher returns, thereby demonstrating the existence of a risk-return trade-off pertaining to a particular risk measure. Since volatility proxies under consideration are likely to be highly correlated, the relation between returns and each proxy is investigated separately, rather than jointly. Lagged observations of the dependent variable are added as regressors to address the potential for temporal dependencies in the return series. The autoregressive order of q is selected based upon the Akaike information criterion (AIC), which indicates that a lag length of one is necessary to generate white noise disturbances.

Apart from volatility, which is intrinsically an internal factor of Bitcoin markets, global factors are likely, in one way or another,

to have a strong bearing on the price dynamics of Bitcoin. [Ciaian et al. \(2016\)](#) point out that changes in equity indices could mirror the overall sentiment of the global economic outlook. Hence, positive (negative) macroeconomic and financial developments are likely to encourage (discourage) the use of Bitcoin in trade and exchanges and thus bolster (weaken) its demand, which eventually could cause Bitcoin prices to increase (decrease). By and large, equity returns may be viewed as an opportunity cost of investing in Bitcoin markets, which implies a positive correlation between equity and Bitcoin prices. Moreover, [Hooker \(2002\)](#) and [Baffes \(2007\)](#) document that oil price spikes contribute substantially to inflationary pressures. In this way, oil price changes could elicit a decrease or an increase in Bitcoin prices. [Baek and Elbeck \(2015\)](#) examine whether Bitcoin returns are driven by a host of fundamental variables, including the S&P 500 index, Euro exchange rates, personal consumption expenditures, 10-year Treasury note, and the national average unemployment rate. [Bouri, Azzi et al. \(2017\)](#) find that the US implied volatility index (VIX) has a significant negative impact on Bitcoin volatility prior to the Bitcoin price crash of December 2013. [Guesmi et al. \(2019\)](#) report evidence of returns and volatility transmission effects between Bitcoin on the one hand and a set of financial assets (equities, gold, oil, fiat currencies) on the other. The results of [Dyhrberg \(2016\)](#) indicate that current returns on Bitcoin are sensitive to past variations in federal funds rate, Financial Times Stock Exchange Index (FTSE Index), and exchange rates of the US dollar against the Euro and British pound. [Ciaian, Rajcaniova, and Kancs \(2018\)](#) document that gold prices, exchange rates of the US dollar vis-à-vis the Euro and Chinese Yuan, and 10-Year Treasury constant maturity rate exert a statistically significant effect on Bitcoin prices, whether in the short or long run.

Accordingly, to control for the potential impact of global influences on the dynamics of Bitcoin returns behavior, Eq. (16) is augmented with a range of factors common to the world financial system

$$r_t = \alpha + \sum_{i=1}^q \xi_i r_{t-i} + \beta v_t + \sum_{l=1}^5 \vartheta_l \wp_{t,l} + \varepsilon_t \quad (17)$$

where $\wp_{t,l}$ is a set of control variables comprising the S&P global broad market index (BMI), the S&P GSCI gold index, the S&P GSCI energy commodities index, the VIX index, and the US trade-weighted exchange rate index.² The inclusion of controls is meant to guarantee that the results from Eq. (16) are not induced by model misspecification, as suggested by [Merton \(1973\)](#); [Scruggs \(1998\)](#); [Bali and Peng \(2006\)](#); [Darrat, Gilley, Li, and Wu \(2011\)](#), among others. The selection of such controls is motivated by empirical findings identifying these factors as relevant sources of price variation in Bitcoin and financial markets (e.g., [Ahmed, 2017](#); [Balcilar, Demir, & Hammoudeh, 2014](#); [Ciaian et al., 2018](#); [Diaz, Molero, & de Gracia, 2016](#); [Dyhrberg, 2016](#); [Johnson & Soenen, 2009](#); [Mensi, Hammoudeh, Yoon, & Balcilar, 2017](#); [Mobarek, Muradoglu, Mollah, & Hou, 2016](#); [Smaoui, Grandes, & Akindele, 2017](#); [Virk, 2012](#)).

In a similar fashion, to explore the intertemporal relation between risk and returns of Bitcoin market, the following regression model is estimated:

$$R_t = \alpha + \sum_{i=1}^q \xi_i R_{t-i} + \beta v_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim (0, \sigma_t^2) \quad (18)$$

where v_{t-1} represents one-day lagged realized volatility measures.

Furthermore, to neutralize the likely influence of global factors on the intertemporal risk-return trade-off, Eq. (18) is extended to incorporate the same array of control variables lagged one period as follows

$$R_t = \alpha + \sum_{i=1}^q \xi_i R_{t-i} + \beta v_{t-1} + \sum_{l=1}^5 \vartheta_l \wp_{t-1,l} + \varepsilon_t \quad (19)$$

As for the estimation approach, since there could be endogeneity between the dependent variable and some of the explanatory variables, the Ordinary Least Squares (OLS) estimates may be biased and inconsistent. To tackle the issue of endogeneity, all regressions are estimated using the GMM approach. A salient characteristic of this estimator is that, unlike other counterparts, it does not impose any restriction on the distribution of the data. The GMM procedure dispenses with the strict exogeneity assumption of the instrumental variables, thus offering flexibility in choosing the appropriate set of instruments. In this study, the instrumental variables considered are a constant, higher order lags of Bitcoin returns, and all relevant covariates with different lag orders.

4. Data and exploratory analysis

4.1. Data description

The main dataset consists of the exchange rate of Bitcoin (Bitstamp exchange) against the US dollar, sampled at an hourly frequency due to data availability limitations. Like foreign currencies, Bitcoins trade 24 h a day, 7 days a week, and 365 days a year. Hence, the intraday hourly prices for Bitcoins span the time interval from 00:00 to 23:00, providing $M = 24$ non-overlapping exchange rate observations per day. The period of analysis extends from January 3, 2012 to April 5, 2019, yielding a sample size of 1894 daily observations. Data for earlier periods are associated with too many inactive trading hours, which dictates the beginning date of the sample. Publicly available online, the price series is retrieved from www.Bitcoincharts.com, the world's premier provider of Bitcoin transaction data on most popular cryptocurrency exchanges such as Bitstamp, Coinbase, Bitfinex, Kraken, and Bitflyer.

² For a brief description of these variables, see Subsection (4.1).

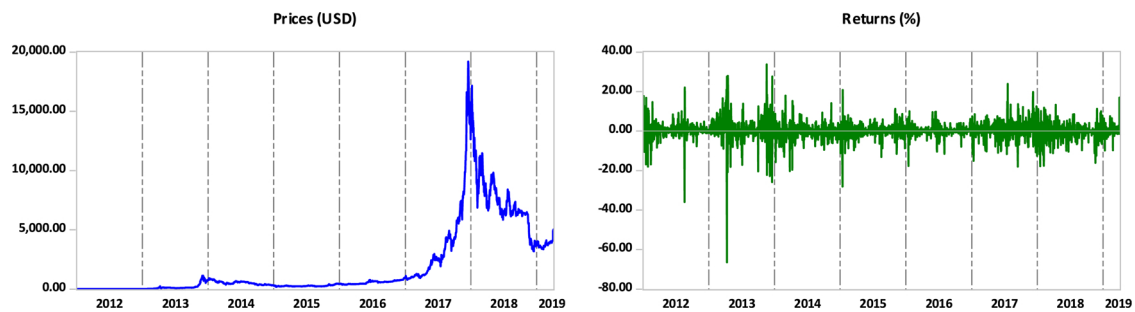


Fig. 1. Price levels and returns of Bitcoin over the sample period.

Hourly returns are calculated as the difference in logarithm between two consecutive prices.

Fig. 1 illustrates both the historical levels and returns of Bitcoin over the entire study period. A closer look at the time evolution of Bitcoin price path indicates that the behavior pattern of upward price spikes followed by mini crashes turns out to be repeated ad nauseam. Particularly, the value of Bitcoin soared to record highs of about \$1150 in 2013, but plunged swiftly later in the same year, and then oscillated between \$180 and \$950 over the course of 2014–2015. The market for Bitcoin started to pick up in 2016, witnessed a phenomenal growth with an extraordinary price surge to around \$20,000 in 2017, before falling prey to a flash crash in early 2018. On the other hand, the returns series seems to exhibit wild swings, where the jump amplitude of Bitcoin price changes varies extremely over time.

With respect to control variables, daily closing price data for the S&P global broad market index (BMI), the S&P GSCI gold index, the S&P GSCI energy commodities index are sourced from S&P Dow Jones Indices database, while those of VIX and US dollar trade-weighted exchange rate indices are obtained from St. Louis Federal Reserve Bank's FRED database. The US trade-weighted exchange rate index is a weighted average value of the US dollar relative to a broad basket of currencies of major US trading partners, gauging the overall strength of the US currency. The S&P global BMI is a comprehensive, rules-based index designed to measure global stock market performance, accommodating about 10,000 stocks from 25 developed and 23 emerging markets. The index covers all publicly listed stocks with a minimum float-adjusted market capitalization of USD 100 million and annual value traded of at least USD 50 million in all involved countries. The S&P GSCI gold and energy indices are extensively utilized commodity index benchmarks, originally recognized as the Goldman Sachs Commodity Indices (GSCI) and later maintained and published by Standard & Poor's. The gold index tracks the COMEX gold future, while the energy index reflects price changes in a basket of energy commodities, including heating oil, unleaded gas, crude oil, and natural gas. The VIX Index is a barometer of market uncertainty and investor sentiment in the US markets, drawing on real-time prices of options on the S&P500. Higher values of VIX reflect investors' fear, anxiety, and downbeat attitude about the near-term direction of market volatility, whereas lower values of VIX are a manifestation of investors' confidence and upbeat sentiment.

It is worth noticing that prior to estimating Eqs. (17) and (19), the time series of realized volatility measures are filtered so that only daily observations in common with those of the control covariates are employed. This calendar alignment is necessary, given that control variables data are available on the usual five trading days of the week. All control variables are expressed as the log first-difference between two successive observations.

4.2. Preliminary analysis

To provide some insight into the stochastic characteristics of the variables of interest, Table 1 reports estimates of the first four unconditional moments, along with normality and unit root test results for each variable. Some salient remarks are worth mentioning.

First, Bitcoin shows by far the largest sample mean returns, compared to the other investment alternatives, namely gold, energy commodities, and world stock markets. Second, the sample average of RV is, by construction, greater than those of the other realized volatility measures. Third, both gold and energy commodity indices exhibit negative mean returns, highlighting the downward price trend in these markets throughout most of the study period. Fourth, VIX and Bitcoin returns experience the highest level of volatility, as shown by their respective standard deviations. Fifth, without exception, the individual series seem to depart considerably from the standard Gaussian distribution. The skewness of the return distributions of Bitcoin, stock and gold markets is negative, suggesting a high probability of substantial losses, rather than gains, in these markets. The kurtosis of each series is materially above three, which implies leptokurtic (fat-tailed) distributions with many extreme values. Of particular note, the excessively high kurtosis statistics for realized volatility measures suggest that wild gyrations in Bitcoin value tend to occur more frequently. As substantiating evidence, the Jarque-Bera test statistics reveal that the null hypothesis of normality is rejected for all series at the 0.01 level of significance. Sixth, the results in the last two columns indicate that the null hypothesis of a unit root is rejected using the Augmented Dickey–Fuller (ADF) test (Dickey and Fuller, 1979), while the KPSS test (Kwiatkowski, Phillips, Schmidt, & Shin, 1992) fails to reject the null hypothesis of stationarity for all series at the 0.01 level. Altogether, these findings confirm that the variables under scrutiny are not in need of further differencing and can be directly modeled in the regression analysis.

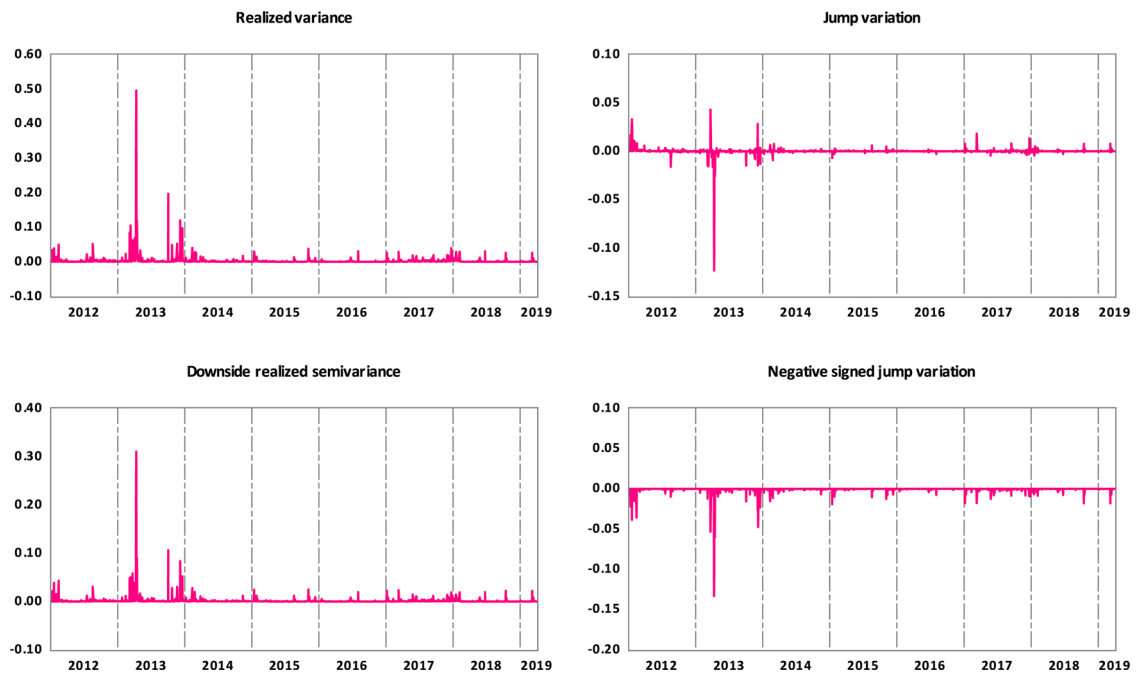
Fig. 2 presents a visual illustration of the daily estimates of realized volatility measures considered in this study. It is quite evident

Table 1

Summary statistics and unit root test results for all variables.

					Unit root tests		
					J-B test	ADF test	KPSS test
	Mean	Standard deviation	Skewness	Kurtosis			
Panel A: Return and realized volatility measures							
Ret	0.462	4.891	−2.053	44.155	1.13E+5***	−32.516***	0.175
RV	3.80E−3	0.019	18.634	430.535	1.21E+7***	−13.521***	0.133
JV	3.79E−4	1.89E−3	14.524	267.569	4.62E+6***	−36.687***	0.038
RS [−]	2.09E−3	0.012	19.202	448.611	1.32E+7***	−14.208***	0.118
SJV [−]	−8.38E−4	5.63E−3	−17.115	357.965	8.29E+6***	−17.354***	0.085
Panel B: Control variables							
BMI	0.032	0.669	−0.663	7.407	1.41E+3***	−32.943***	0.061
GLD	−0.012	0.977	−0.543	11.413	4.77E+3***	−40.811***	0.037
ENR	−0.033	1.707	0.137	6.255	708.439***	−42.749***	0.112
VIX	0.015	7.495	1.295	13.032	7.13E+3***	−42.092***	0.044
USX	0.009	0.286	0.063	7.072	1.10E+3***	−39.128***	0.127

Notes: This table presents some univariate properties of the series under investigation over the sample period. Ret is the daily Bitcoin's rate of return. RV, JV, RS⁻, and SJV⁻ are proxy measures of the intraday price variability, including realized variance, jump variation, downside realized semivariance, and negative signed jump variation, respectively. BMI, GLD, ENR, VIX, and USX are a set of control variables, including the S&P global broad market index, the S&P GSCI gold index, the S&P GSCI energy commodities index, the Chicago Board Options Exchange volatility index, and the US trade-weighted exchange rate index, respectively. J-B represents the Jarque-Bera test statistic for the null hypothesis of normality. ADF and KPSS are the Augmented Dickey-Fuller test and the Kwiatkowski, Phillips, Schmidt, and Shin test for the null hypothesis of nonstationarity and stationarity of a univariate time series, respectively. Sourced from [MacKinnon \(1996\)](#), the critical value for the ADF test, with a constant and a deterministic trend, is -3.964 at the 0.01 level of significance. Obtained from [Kwiatkowski et al. \(1992\)](#), the critical value for the KPSS test, with a constant and a deterministic trend, is 0.216 at the 0.01 level of significance. *** indicates rejection of the corresponding null hypothesis at the 0.01 significance level.

**Fig. 2.** Daily realized volatility proxy measures of Bitcoin.

that RV, JV, RS⁻ tend to exhibit a common evolutionary pattern over time, which implies that when RV is large in magnitude, both JV and RS⁻ are also high, and vice versa. Furthermore, SJV⁻ appears to be much more clustered in the period 2012–2013 and in 2015, reflecting dramatic falls in Bitcoin value during these periods.

Table 2

GMM estimation results for the contemporaneous risk-return relation without control variables.

Realized volatility measures				
	RV	JV	RS ⁻	SJV ⁻
α	0.342* (1.826)	0.022 (0.604)	0.218** (2.506)	0.055** (2.381)
ξ_1	0.398 (1.388)	0.625** (2.176)	0.527*** (3.877)	0.231*** (5.208)
β	-0.479** (-2.287)	-0.373*** (-4.064)	-0.606*** (-7.334)	-0.279** (-2.219)
P-value of Hansen's J-test	0.677	0.396	0.427	0.406
Adjusted R ²	0.339	0.421	0.386	0.351

Notes: This table reports estimates from GMM regressions of daily returns on lagged returns and contemporaneous realized volatility measures. RV, JV, RS⁻, and SJV⁻ are proxy measures of the intraday price variability, including realized variance, jump variation, downside realized semivariance, and negative signed jump variation, respectively. α is an intercept term. ξ_1 is the coefficient estimate on the lagged dependent variable. β represents the estimated coefficient on a particular realized volatility measure. In each regression, the autoregressive lag length is determined using the AIC criterion. The values given in parentheses are heteroskedasticity- and autocorrelation-consistent (HAC) *t*-statistics calculated using the Newey and West (1987) standard errors. Test statistics for individual coefficients examine the null hypothesis that the corresponding coefficient is not different from zero. The Hansen's *J*-statistic test assesses instrument validity/exogeneity, operating under the null hypothesis that the instrumental variables are jointly exogenous. The asterisks ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

5. Empirical evidence

5.1. Contemporaneous risk-return relation

This subsection is concerned with investigating the presence and significance of a contemporaneous relation between realized volatility proxies and returns of Bitcoin.

Table 2 reports the GMM regression results from Eq. (16) that does not include control variables. The estimated first-order autoregressive coefficient, ξ_1 , in all equations but that of RV is positive and statistically different from zero at the 0.05 significance level or better. This result suggests that current price changes of Bitcoin are positively affected by previous price changes, underscoring the importance of taking into account the dynamic nature of this market. In line with this evidence, Baur, Dimpfl, and Kuck (2018) and Dyhrberg (2016) find that past Bitcoin market returns have a positive influence on current returns. Irrespective of the risk measure used, the coefficient estimate of the risk-return relation, β , is negative and statistically significant at the 0.05 level or better. This implies that elevated levels of, JV, RS⁻, and SJV⁻ are contemporaneously associated with price declines. Clearly, in terms of magnitude and statistical significance, RS⁻ appears to have the largest negative impact on Bitcoin returns. Consistent with these results, Eross, McGroarty, Urquhart, and Wolfe (2019), utilizing correlation analysis, find a negative relation between Bitcoin returns and realized volatility.

The next exercise is to reassess the contemporaneous risk-return relation in the presence of a set of global control variables. The GMM parameter estimates of Eq. (17) are shown in Table 3.

Interestingly, the results remain qualitatively unaltered, as the incorporation of such controls makes no noticeable difference with respect to the underlying link between volatility and returns. More explicitly, the autoregressive coefficient estimate in RV regression becomes statistically different from zero at the 0.05 level. In terms of sign and statistical significance, the estimated coefficients on, JV, RS⁻, and SJV⁻ in the full regression model with control variables seem to be fairly similar to their respective counterparts in the baseline model. After introducing the control variables, the negative influence of, JV, and RS⁻ turns out to be slightly stronger, as their respective coefficient estimates increase in magnitude. Contrarily, the SJV⁻ coefficient becomes less negative. These findings are in conflict with those of Bouri, Azzi et al. (2017), who, using an asymmetric-GARCH model, establish that return shocks are positively correlated with volatility before the Bitcoin price crash of 2013, but not after the crash.

As far as control variables are concerned, there is consistent statistical evidence on lack of a contemporaneous association between Bitcoin returns and those of gold, energy, and the US trade-weighted exchange rate, across all regressions. The coefficient estimates related to these global factors look far below standard levels of significance, no matter which realized measure is employed. This result implies that Bitcoin could potentially serve as a hedge asset against price fluctuations in gold and energy markets. Likewise, Polasik, Piotrowska, Wisniewski, Kotkowski, and Lightfoot (2015) and Baur, Dimpfl et al. (2018) demonstrate no significant contemporaneous correlation between Bitcoin returns and the broad trade-weighted dollar index changes. Using wavelet coherence analysis, Kristoufek (2015) reports evidence that Bitcoin prices are not affected by the dynamics of gold. However, Symitsi and Chalvatzis (2018) find significant return spillover effects from energy stocks to Bitcoin. On the other hand, the results from the JV and RS⁻ regressions show that world stock markets, as represented by the S&P global broad market index, have a statistically positive influence on Bitcoin returns at the 0.05 level. Such positive relation suggests that Bitcoin and stocks are likely to fail to hedge each other's risk exposure. In contrast, Baek and Elbeck (2015) and Baur, Hong et al. (2018) document that the S&P 500 index returns are uncorrelated with Bitcoin returns. With respect to the global uncertainty and fear index, the associated coefficient estimate in RV regression bears a negative sign, but it lacks statistical significance. In the remaining cases, however, daily variations in the VIX are

Table 3

GMM estimation results for the contemporaneous risk-return relation with control variables.

Realized volatility measures				
	RV	JV	RS ⁻	SJV ⁻
α	0.534** (2.037)	0.223** (2.265)	0.819*** (6.904)	0.329 (0.747)
ξ_1	0.453** (2.395)	0.175* (1.772)	-0.562** (-2.488)	0.052*** (3.792)
β	-0.522*** (-5.055)	-0.421*** (-2.971)	-0.653** (-2.408)	-0.203** (-2.343)
ϑ_{BMT}	0.386 (0.179)	0.498** (2.311)	0.224** (2.093)	0.673 (0.910)
ϑ_{GLD}	-0.581 (-0.197)	-0.232 (-0.134)	0.022 (0.981)	-0.721 (-1.227)
ϑ_{ENR}	0.430 (0.359)	-0.078 (-0.530)	0.146 (1.541)	-0.049 (-0.095)
ϑ_{VIX}	-0.182 (-0.456)	-0.454** (-2.188)	-0.260*** (-4.009)	-0.342** (-2.207)
ϑ_{USX}	1.643 (0.180)	0.502 (1.099)	0.107 (0.762)	0.334 (0.239)
P-value of Hansen's J-test	0.297	0.219	0.211	0.181
Adjusted R ²	0.411	0.483	0.439	0.408

Notes: This table presents estimates from GMM regressions of daily returns on lagged returns, contemporaneous realized volatility measures, and a set of contemporaneous control variables. RV, JV, RS⁻, and SJV⁻ are proxy measures of the intraday price variability, including realized variance, jump variation, downside realized semivariance, and negative signed jump variation, respectively. α is an intercept term. ξ_1 is the coefficient estimate on the lagged dependent variable. β represents the estimated coefficient on a particular realized volatility measure. ϑ_{BMT} , ϑ_{GLD} , ϑ_{ENR} , ϑ_{VIX} , and ϑ_{USX} are the coefficient estimates on the control variables, including the S&P global broad market index, the S&P GSCI gold index, the S&P GSCI energy commodities index, the Chicago Board Options Exchange volatility index, and the US trade-weighted exchange rate index, respectively. In each regression, the autoregressive lag length is determined using the AIC criterion. The values given in parentheses are heteroskedasticity- and autocorrelation-consistent (HAC) *t*-statistics calculated using the Newey and West (1987) standard errors. The asterisks ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

negatively correlated with Bitcoin returns at the 0.05 level or better.

Finally, in terms of model specification, the last two rows of Tables 2 and 3 show *p*-values of the Hansen's *J*-test for over-identifying restrictions and the adjusted R² for each GMM regression. The diagnostic test results demonstrate the validity and strength of instrumental variables. Without exception, all significance probabilities are uniformly above the 0.10 level, implying that the null hypothesis of orthogonality between the set of instruments adopted in each regression and the corresponding residuals cannot be rejected. Besides, the overall explanatory power of the individual regression models seems to be rather satisfactory. As seen in Table 2, the adjusted R² values range between 0.339 for RV regression and 0.421 for JV regression. With the inclusion of control variables, as shown in Table 3, there is some improvement in the adjusted R² lying between 0.408 for SJV⁻ regression and 0.483 for JV regression.

Table 4

GMM estimation results for the intertemporal risk-return relation without control variables.

Realized volatility measures				
	RV	JV	RS ⁻	SJV ⁻
α	0.260** (2.005)	0.067 (1.459)	0.229** (2.323)	0.215* (1.724)
ξ_1	0.796*** (3.390)	0.301*** (3.103)	0.517** (2.454)	0.978** (2.516)
β_1	-0.331* (-1.689)	-0.253* (-1.711)	-0.653* (-1.739)	-0.427 (-0.763)
P-value of Hansen's J-test	0.249	0.318	0.635	0.355
Adjusted R ²	0.392	0.441	0.426	0.377

Notes: This table reports estimates from GMM regressions of daily returns on lagged returns and lagged realized volatility measures. α is an intercept term. ξ_1 is the coefficient estimate on the lagged dependent variable. β_1 denote the coefficient estimate on the lagged realized volatility measure. In each regression, the autoregressive lag length is determined using the AIC criterion. The values given in parentheses are heteroskedasticity- and autocorrelation-consistent (HAC) *t*-statistics calculated using the Newey and West (1987) standard errors. The asterisks ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

Table 5

GMM estimation results for the intertemporal risk-return relation with control variables.

Realized volatility measures				
	RV	JV	RS [−]	SJV [−]
α	0.705** (2.292)	0.581 (1.444)	0.955 (1.327)	0.043* (1.703)
ξ_1	0.484*** (2.948)	0.263** (2.473)	0.647* (1.884)	0.608*** (5.441)
β_1	−0.489* (−1.695)	−0.329* (−1.757)	−0.742** (−2.285)	−0.255 (−0.903)
$\vartheta_{BML,1}$	0.207* (1.726)	0.709*** (4.091)	0.407** (2.414)	0.399*** (5.643)
$\vartheta_{GLD,1}$	−0.208* (−1.698)	−0.874 (−0.698)	−0.301* (−1.715)	−0.307 (−1.402)
$\vartheta_{ENR,1}$	1.082 (0.665)	−0.046 (−0.801)	0.942 (1.403)	0.177 (0.603)
$\vartheta_{VIX,1}$	−0.398*** (−5.098)	−0.532** (−2.383)	−0.272*** (−6.036)	−0.497** (−2.466)
$\vartheta_{USX,1}$	−0.779 (−1.205)	0.140 (1.075)	−0.657 (−0.813)	0.839 (1.227)
P-value of Hansen's J-test	0.201	0.353	0.475	0.257
Adjusted R ²	0.445	0.471	0.511	0.435

Notes: This table presents estimates from GMM regressions of daily returns on lagged returns, one-day lagged realized volatility measures, and a set of one-day lagged control variables. α is an intercept term. ξ_1 is the coefficient estimate on the lagged dependent variable. β_1 denote the coefficient estimate on the lagged realized volatility measure. $\vartheta_{BML,1}$, $\vartheta_{GLD,1}$, $\vartheta_{ENR,1}$, $\vartheta_{VIX,1}$, and $\vartheta_{USX,1}$ are the coefficient estimates on the one-period lagged control variables, including the S&P global broad market index, the S&P GSCI gold index, the S&P GSCI energy commodities index, the Chicago Board Options Exchange volatility index, and the US trade-weighted exchange rate index, respectively. In each regression, the autoregressive lag length is determined using the AIC criterion. The values given in parentheses are heteroskedasticity- and autocorrelation-consistent (HAC) *t*-statistics calculated using the Newey and West (1987) standard errors. The asterisks ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

5.2. Intertemporal risk-return relation

In this subsection, the intertemporal risk-return relation is examined. The baseline regression results of Eq. (18) are presented in Table 4.

Across all regressions, the coefficient on the lagged dependent variable is still positive and statistically differs from zero at the 0.05 or 0.01 level, confirming that the information content of past price changes in Bitcoin has substantial predictive power for current price changes. In *RV*, *JV*, and *RS[−]* regressions, the estimated coefficient of lagged volatility, β_1 , carries a negative sign, with a borderline statistical significance ($p \geq 0.082$). Accordingly, the intertemporal relation between Bitcoin returns and each of realized variance, jump variation, and downside realized semivariance is negative, but weak in a statistical sense. In *SJV[−]* regression, β_1 is also negatively signed, yet it fails to attain traditional levels of statistical significance. In terms of magnitude, the negative effect of *RS[−]* on returns seems to outweigh those of the other realized measures.

Next, the augmented regression results of Eq. (19) are shown in Table 5. To a great extent, the inclusion of control variables does not markedly alter the nature and strength of the intertemporal risk-return relation. More explicitly, all coefficient estimates of *JV*, *RS[−]*, and *SJV[−]* retain their negative sign, confirming the negative effect of lagged realized volatility measures on current returns. While the statistical significance of *RV* and *JV* is still marginal, it becomes stronger for *RS[−]*. Once again, the coefficient estimate on *SJV[−]* is not statistically different from zero. The intertemporal influence of *JV*, and *RS[−]* on returns increases somewhat in magnitude, following the entry of the control variables.

Regarding control variables, the results reveal that lagged values of the S&P global broad market index and the VIX are important determinants of the variation in Bitcoin prices. In all regressions, the representative index of world stock markets appears to have a positive effect on Bitcoin returns, as indicated by the statistical significance of its corresponding estimated coefficients at conventional levels. This effect is most (least) obvious in *JV* (*RV*) regression. Likewise, the representative index of global uncertainty, VIX, is negatively correlated with Bitcoin returns across all regressions, as shown by the statistical significance of its corresponding coefficients at the 0.05 or 0.01 level. The size of the effect is most pronounced when *JV* is used as a proxy for volatility. Lagged price changes in gold are negatively associated with Bitcoin returns, but with a marginal statistical significance in *RV* and *RS[−]* regressions. The coefficients on energy commodity index and the US trade-weighted exchange rate index are statistically indistinguishable from zero, regardless of the volatility measure deployed. This finding confirms the irrelevance of either factor to explaining movements of Bitcoin prices.

The last two rows of Tables 4 and 5 provide *p*-values of the Hansen's *J*-test and the adjusted *R*². The null hypothesis that the instrumental variables are uncorrelated with the error terms cannot be rejected in all cases, as the significance probabilities are unequivocally higher than the 0.10 level. The overall explanatory power of the GMM models is still reasonable. In Table 4, the adjusted *R*² values lie between 0.377 for *SJV[−]* regression and 0.441 for *JV* regression. The introduction of control variables, however,

brings a slight improvement in the explanatory power of each regression model. As shown in Table 5, the adjusted R^2 values range between 0.435 for SJV^- regression and 0.511 for RS^- regression.

5.3. Possible explanations

In brief, this study presents evidence of a negative risk-return relation for Bitcoin markets, whether at the contemporaneous or intertemporal level, which implies that higher levels of volatility tend to depress returns, whereas lower volatility levels increase returns. Although counterintuitive, this inverse association is not uncommon in international financial markets. As a vivid example, a large stream of the relevant stock-market literature suggests that returns are negatively correlated with volatility (e.g., Aslanidis, Christiansen, & Savva, 2016; Brandt & Kang, 2004; Jin, 2017; Li, 2011; Peterson & Smedema, 2011; Whitelaw, 1994, among others). In a comprehensive review of the evidence on the low risk effect, Blitz, Vliet, and Guido (2019) conclude that the low risk anomaly is consistently present over time, and across firms, industries, and countries. As discussed in the Introductory section, the empirical failure of the positive risk-return trade-off has spurred a growing research attempting to offer reasonable justifications for this anomaly. Some of the behavioral finance-based explanations set forth in the stock market context appear to have much relevance to Bitcoin markets.

In this respect, one of the most pertinent explanations is the reference-dependent preference theory (RDPT), developed by Wang et al. (2017), which is based upon the prospect theory (PT) of Kahneman and Tversky (1979). A principal assumption of standard asset pricing models is that individuals are consistently risk-averse, which connotes that they have a concave utility function for their wealth. In PT, however, payoffs that individuals obtain are defined in terms of gains or losses, which are evaluated against a particular reference or anchor point. Since people naturally tend to be more affected by losses than by comparable gains, their value function takes an asymmetric S-shape, wherein gains (losses) have a concave (steeper convex) curve. Accordingly, as noted by Berejikian (2002), when it comes to choosing between competing gains (losses) relative to the reference point, most individuals show risk-averting (risk-seeking) behavior. This means that, in many instances, a single investor could display different risk attitudes, which, in turn, rest on which domain (i.e., gains or losses) s/he is in. Within the framework of RDPT, since investors fear the potential for losses, they are often predisposed towards holding onto losing assets (e.g., stocks, cryptocurrencies) longer than they should, in the hope of recovering their invested capital. Alternatively, investors may prefer to lock in gains by selling winning assets, in order to avoid any future losses. Shefrin and Statman (1985) couch such a pattern of behavior as disposition effect, in which investors rush to sell well-performing assets too quickly but defer selling poor-performing ones too long. Hence, as predicted by Wang et al.'s (2017) theory, investors, possessing assets with current market prices below their corresponding reference price levels, are inclined to adopt risk-seeking behavior as they incur capital losses. For this loser-asset class, therefore, there should be an inverted risk-return relation, especially if there are shorting and leverage constraints. Quite the contrary, investors, who have assets with market values above their corresponding reference levels, engage in risk-avoidance behavior as they accomplish capital gains. Thus, a positive risk-return trade-off for the winner-asset class should be present.

Other potential explanations draw upon psychological phenomena, such as cognitive biases that subconsciously can mold individuals' perceptions and decisions, stressing that not all investors are entirely rational, as standard asset pricing theory assumes. Barberis and Huang (2008); Baker et al. (2011), and Hsu and Chen (2017), among others, propose that the low risk anomaly may emerge from a preference for positive skewness or, equivalently, lottery-like payoffs. Many retail investors with strong gambling propensities usually gravitate towards lottery-type investments (e.g., Bitcoin), which offer limited downside risk but unlimited upside potential. Kumar (2009) argue that individual investors perceive low-priced assets with high idiosyncratic volatility and high idiosyncratic skewness as valid gambling vehicles. Blitz et al. (2019) suggest that a preference for lottery-like, or positively skewed, wealth distributions can explain the anomalous negative risk-return association. That is, rather than commanding a risk compensation for holding lottery-type assets, investors with a gambling mindset are willing to pay a premium to benefit from the upside potential of these assets. Based on the cumulative prospect theory of Tversky and Kahneman (1992); Barberis and Huang (2008) document that an asset's idiosyncratic skewness can be a priced factor. The authors indicate that when agents have the cumulative prospect theory preferences, assets with positive skewness can be overpriced and yield negative average excess returns. Mitton and Vorkink (2007) demonstrate that investors with lottery-like preferences tend to sacrifice mean-variance efficiency so as to boost the probability of extreme positive returns or, equivalently, to obtain greater levels of positive skewness in their portfolios.

Hong and Sraer (2016) develop an alternative explanation, which posits that heterogeneity of investors' expectations about the market or common factor of companies' cash flows combined with short-sales constraints can bring about overvaluation, and especially so for high-beta (risk) assets. Aggregate disagreement emanates from contradictory priors (beliefs) or psychological biases (e.g., overreaction, overconfidence, representativeness). In their model, the authors contend that the speculative nature of high-beta assets dominates the risk-sharing effect and the expected returns of such assets can be lower than those of low-beta ones. Given that high-beta assets are associated with large loadings on the market factor, investors disagree more on the payoffs of these assets when divergence of opinion on the market factor is intense. Most likely, the value of high-beta assets is determined by optimistic investors who maintain an upbeat view of the market factor. Arbitrageurs, on the other side, fail to completely correct the mispricing due to their risk-aversion. Ultimately, this leads to lower subsequent returns for high-beta assets compared to their low-beta counterparts.

The volatility feedback hypothesis (Campbell & Hentschel, 1992; Pindyck, 1984; Poterba & Summers, 1986) could possibly provide a further explanation for the volatility-driven negative correlation with returns. Such a hypothesis postulates that the arrival of new information, whether favorable or unfavorable, to the market heralds an increase in the volatility level and, consequently, the risk-adjusted discount rates. Climbing risk premiums partly offset the positive effect of favorable information on stock prices, while the negative effect of unfavorable information on prices tends to intensify. Under the assumption that future cash flows are

unaffected, either case implies an immediate price decline.

6. Robustness analysis

To evaluate the robustness of the findings reported so far, three investigations are carried out in this section. More explicitly, first, a consolidated specification is developed, which accommodates both contemporaneous and lagged effects of Bitcoin volatility. This exercise aims to examine how sensitive are returns to both current and lagged volatilities. Second, to ascertain whether the results are still valid beyond the Bitcoin-market sphere, the analysis is replicated utilizing the second top-ranked digital currency, Ethereum. Third, the relation between Bitcoin returns and realized measures of risk beyond the second-order moment of the underlying return distribution (i.e., realized skewness and realized kurtosis) is also explored. This helps to determine if higher-order risk measures matter for the risk-return relation. To conserve space, the following discussions are limited to the main regressors of interest, i.e., measures of realized risk.³

6.1. Concurrent and lagged volatility effects

Hitherto, the analysis has considered separately the contemporaneous and intertemporal risk-return relations. One may next wish to assess the relative importance of current and lagged volatility in explaining current returns, after controlling for a set of influential determinants of Bitcoin's price formation process. In a similar spirit to Aslanidis et al. (2016) and Ludvigson and Ng (2007), among others, an alternative multivariate specification is estimated, wherein current returns of Bitcoin are expressed as a function of their own lags, current and lagged values of volatility measures, along with current values of the control variables. Accordingly, Eq. (17) is modified as follows:

$$R_t = \alpha + \sum_{i=1}^q \xi_i R_{t-i} + \beta_1 \mathcal{V}_t + \beta_2 \mathcal{V}_{t-1} + \sum_{l=1}^5 \vartheta_l \mathcal{Q}_{t,l} + \varepsilon_t \quad (20)$$

The above equation is estimated individually for each measure of volatility, namely, JV , RS^- , and SJV^- .⁴ The AIC criterion suggests a lag order of two (one) for the RV and SJV^- (and RS^-) regressions, to obtain white noise residuals. Table 6 reports the estimation results.

As seen in Table 6, the coefficient estimates of the contemporaneous risk-return relation, β_1 , are still negative in sign and distinguishable from zero at the 0.05 significance level or better, no matter which volatility measure is utilized. As is the case in Tables 2 and 3, RS^- appears to have the most pronounced negative influence on returns. On the other hand, the lagged coefficient estimates of RV , JV , and RS^- preserve their negative sign, but they are somewhat weaker, in terms of magnitude and statistical significance, relative to their corresponding contemporaneous measures. In SJV^- regression, β_2 is still negatively signed, even though it decreases in size and loses statistical significance. It is obvious that the lagged negative impact of JV outweighs those of the other volatility measures. In general, these results lead to the conclusion that Bitcoin returns seem to be more sensitive to current volatility than to lagged volatility.

6.2. The case of Ethereum

Despite the fact that virtual currencies, in general, share the basic fundamentals of blockchain technology and reward mechanism, they may exhibit different stochastic behaviors. The results of Omane-Adjepong, Alagidede, and Akosah (2019) show heterogeneous long-term dependence characteristics for eight major cryptocurrency markets. Using daily data of 17 virtual currencies, Ciaian et al. (2018) find that only 4 altcoins (i.e., Ethereum, Namecoin, Nxt, and SuperNET) have a long-run relation with Bitcoin. Consequently, it is important to verify whether the evidence documented in this study is Bitcoin-specific or indicative of a broad pattern in digital currency markets. For this purpose, the analysis is extended to consider the risk-return relation of Ethereum, the world's second largest cryptocurrency by total market capitalization (<https://coinmarketcap.com>; accessed on September 14, 2019). Ethereum's hourly exchange rates relative to the US dollar are collected for the period from May 9, 2016 to April 5, 2019. The starting date of the series is constrained by data availability.⁵ Next, Eq. (20) is re-estimated, such that Ethereum's current return is modeled as a function of its own lags, current and lagged volatility measures, together with current values of the control variables. The GMM parameter estimates are provided in Table 7.

The results in Table 7 sketch a rather similar picture to those documented for Bitcoin. At the contemporaneous level, Ethereum returns are inversely correlated with volatility, regardless of the realized measure deployed. Nonetheless, the evidence is statistically weak in case of RV , JV , and RS^- regressions, and insignificant in case of SJV^- regression. At the intertemporal level, the relation is more pronounced, with the corresponding coefficients, β_2 , becoming stronger in terms of magnitude and statistical significance. The negative influence of RS^- tends to prevail over those of the other volatility measures, both contemporaneously and intertemporally. Altogether, these findings confirm the negative risk-return associations in Bitcoin and Ethereum markets. It should yet be noticed

³ The author would like to thank the anonymous reviewers for suggesting this battery of robustness checks.

⁴ Specifications that include one-period lagged values of control variables are also examined. To a great extent, the results remain qualitatively unaltered.

⁵ The exchange rate series is sourced from <https://www.cryptodatadownload.com/data/northamerican/>.

Table 6
GMM estimates of the risk-return relations.

Realized volatility measures				
	RV	JV	RS [−]	SJV [−]
α	0.429* (1.834)	0.329** (2.211)	0.612* (1.707)	0.291 (0.316)
ξ_1	0.503** (2.083)	0.497** (2.141)	0.255*** (3.343)	0.475*** (4.391)
ξ_2	0.266** (2.264)	—	—	0.362* (1.799)
β_1	−0.431** (−2.306)	−0.416** (−2.403)	−0.506*** (−4.832)	−0.385** (−2.209)
β_2	−0.306* (−1.782)	−0.386** (−2.136)	−0.264** (−2.397)	−0.297 (−1.333)
ϑ_{BMI}	0.325** (2.432)	0.548** (2.238)	0.325* (1.788)	0.453** (2.003)
ϑ_{GLD}	−0.198*** (−3.840)	−0.463* (−1.874)	−0.532* (−1.887)	−0.239* (−1.821)
ϑ_{ENR}	0.547 (0.886)	−0.135* (−1.833)	−0.230* (−1.765)	0.458 (1.367)
ϑ_{VIX}	−0.538** (−2.445)	−0.232* (−1.773)	−0.287** (−2.064)	−0.543*** (−4.578)
ϑ_{USX}	−0.271* (−1.907)	0.264 (0.548)	0.353 (1.461)	0.542 (1.369)
<i>P</i> -value of Hansen's <i>J</i> -test	0.137	0.231	0.273	0.175
Adjusted <i>R</i> ²	0.566	0.526	0.543	0.466

Notes: This table presents estimates from GMM regressions of daily returns on lagged returns, contemporaneous and one-day lagged realized volatility measures, and a set of contemporaneous control variables. α is an intercept term. ξ_1 and ξ_2 are the coefficient estimates on the one- and two-day lagged dependent variable, respectively. β_1 and β_2 denote the coefficient estimates of the contemporaneous and intertemporal risk-return relations, respectively. In each regression, the autoregressive lag length is determined using the AIC criterion. The values in parentheses are heteroskedasticity- and autocorrelation-consistent (HAC) *t*-statistics calculated using the Newey and West (1987) standard errors. The asterisks ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

Table 7
GMM estimates of the risk-return relations for Ethereum.

Realized volatility measures				
	RV	JV	RS [−]	SJV [−]
α	0.054 (1.434)	0.098* (1.908)	0.119** (2.207)	0.358** (2.408)
ξ_1	0.383** (2.421)	0.276* (1.755)	0.219** (2.086)	0.306** (2.218)
ξ_2	0.165* (1.740)	0.205 (1.339)	0.156* (1.751)	0.287** (2.372)
β_1	−0.265* (−1.833)	−0.329* (−1.925)	−0.443* (−1.768)	−0.310 (−1.541)
β_2	−0.373*** (−6.860)	−0.427** (−2.355)	−0.496*** (−3.997)	−0.341** (−2.209)
ϑ_{BMI}	0.195*** (4.008)	0.353* (1.871)	0.468** (2.074)	0.272** (2.422)
ϑ_{GLD}	−0.387 (−0.984)	−0.310 (−1.353)	−0.462* (−1.750)	−0.456 (−0.753)
ϑ_{ENR}	−0.228* (−1.753)	−0.277** (−2.313)	−0.304** (−2.264)	0.343 (1.267)
ϑ_{VIX}	−0.219** (−2.076)	−0.483 (−0.845)	−0.392** (−2.131)	−0.479** (−2.256)
ϑ_{USX}	−0.481* (−1.865)	0.323 (1.158)	−0.264* (−1.896)	0.654 (1.531)
<i>P</i> -value of Hansen's <i>J</i> -test	0.163	0.385	0.519	0.284
Adjusted <i>R</i> ²	0.483	0.422	0.512	0.416

Notes: This table presents estimates from GMM regressions of Ethereum returns on lagged returns, contemporaneous and one-day lagged realized volatility measures, and a set of contemporaneous control variables. α is an intercept term. ξ_1 and ξ_2 are the coefficient estimates on the one- and two-day lagged dependent variable, respectively. β_1 and β_2 denote the coefficient estimates of the contemporaneous and intertemporal risk-return relations, respectively. In each regression, the autoregressive lag length is determined using the AIC criterion. The values in parentheses are heteroskedasticity- and autocorrelation-consistent (HAC) *t*-statistics calculated using the Newey and West (1987) standard errors. The asterisks ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

that, for Bitcoin (Ethereum), the evidence seems to be more robust, in terms of statistical significance, at the contemporaneous (intertemporal) level than at the intertemporal (contemporaneous) level.

6.3. Bitcoin returns and higher realized moments

One of the primary assumptions of the modern portfolio theory of Markowitz (1952) is that asset returns follow a Gaussian distribution, which can be fully characterized by the first moment (mean) and second moment (variance). However, when returns exhibit asymmetry and heavy tails, as is the case with Bitcoin's (see Table 1), it becomes important to consider higher-order moments (i.e., skewness and kurtosis) as more adequate indicators of variability and dispersion. As far as theory is concerned, Arditti (1967); Rubinstein (1973), and Kraus and Litzenberger (1976), among others, suggest that higher-order risks are relevant for explaining returns on risky assets. A related strand of research proposes three- and four-parameter capital asset pricing models, in order to account for the effects of skewness and kurtosis on security returns (e.g., Diacogiannis, 1994; Doan, Lin, & Zurbrugg, 2010; Fang & Lai, 1997; Kraus & Litzenberger, 1976; Neuberger, 2012). Results of a wide range of studies show that higher moments of the return distribution do matter for equity pricing purposes (e.g., Chang, Christoffersen, & Jacobs, 2013; Conrad, Dittmar, & Ghysels, 2013; Dittmar, 2002; Harvey & Siddique, 2000). Kinatder and Papavassiliou (2019) establish that realized kurtosis is the most important driver of subsequent sovereign bond returns for GIPS countries (i.e., Greece, Italy, Portugal and Spain) during turbulent periods. Employing data from 10 foreign exchange markets, Christiansen (2011) finds that realized volatility (realized skewness) has a contemporaneous significant negative (positive) impact on excess returns for all currencies examined.

That being so, it is of great interest to explore whether Bitcoin returns are also sensitive to higher-order risk factors, given their prominent role in explaining returns of various asset classes. To this end, the following non-parametric model-free measurements of realized skewness, RSK_t , and kurtosis, RKU_t , introduced by Amaya, Christoffersen, Jacobs, and Vasquez (2015), are utilized in the analysis:

$$RSK_t = M^{1/2} \cdot \sum_{i=1}^M R_{t,i}^3 / RV_t^{3/2} \quad (21)$$

$$RKU_t = M \cdot \sum_{i=1}^M R_{t,i}^4 / RV_t^2 \quad (22)$$

As indicated in Subsection 3.1, M is the total number of observations of intraday returns, $R_{t,i}$ represents the intraday return i within a trading day t where $i = 1, 2, \dots, M$, and RV is the daily realized variance. To uncover the linkages between returns and higher-order moments, at contemporaneous and intertemporal levels, the following regressions are estimated:

$$R_t = \alpha + \sum_{i=1}^q \xi_i R_{t-i} + \beta_1 RSK_t + \beta_2 RSK_{t-1} + \sum_{l=1}^5 \vartheta_l \varphi_{t,l} + \varepsilon_t \quad (23)$$

$$R_t = \alpha + \sum_{i=1}^q \xi_i R_{t-i} + \beta_1 RKU_t + \beta_2 RKU_{t-1} + \sum_{l=1}^5 \vartheta_l \varphi_{t,l} + \varepsilon_t \quad (24)$$

The GMM parameter estimates of Eqs. (23) and (24) are listed in Table 8. As shown in column 1 of Table 8, the coefficient estimates on the realized third moment of returns, β_1 and β_2 , are negatively signed and statistically distinguishable from zero at the 0.10 level or better, confirming the negative nexus between realized skewness and both contemporaneous and subsequent returns. Nevertheless, the magnitude of either effect is rather modest (i.e., -0.097 and -0.116). This finding is comparable to those of past works (e.g., Amaya et al., 2015; Barberis & Huang, 2008; Kraus & Litzenberger, 1976; Mitton & Vorkink, 2007; Xing, Zhang, & Zhao, 2010), which report a negative relation between skewness and returns in equity markets. Arditti (1967) considers negative skewness as a reasonably further surrogate for the riskiness of individual stocks. A plausible interpretation of the negative sign associated with the third moment measure is that since negative skewness implies a higher probability of extreme losses, investors engaged with Bitcoin markets expect to earn higher returns as a compensation for taking on downside risk. Kumar (2009) demonstrates that individual investors, characterized by specific attributes, have a strong predilection for lottery-like assets, i.e., stocks with lower mean returns, higher idiosyncratic volatility, higher skewness, and lower prices.

On the other hand, the results in column 2 of Table 8 show that the estimated coefficients on the realized fourth moment of returns, β_1 and β_2 , are positive in sign but statistically indifferent from zero at standard levels. The magnitude of either effect appears to be too small to be material (i.e., 0.008 and 0.021). This evidence suggests that excess kurtosis has no bearing on Bitcoin's current and future returns. As a consequence, one can conclude that realized skewness, but not kurtosis, is an important determinant of returns Bitcoin returns. In a similar spirit, Bai, Bali, and Wen (2019) document that kurtosis makes a marginal incremental contribution, in terms of economic magnitude and statistical significance, to the downside risk premium of corporate bonds. Based upon data from 13 European stock markets, Aslanidis et al. (2016) find that both skewness and kurtosis have no significant role in explaining the risk-return trade-off.

Table 8
The relation between Bitcoin returns and higher realized moments.

	(1) Realized skewness	(2) Realized kurtosis
α	−0.187 (−0.754)	0.280* (1.798)
ξ_1	0.336*** (5.095)	0.271* (1.778)
ξ_2	0.208** (2.233)	0.315** (2.501)
$\mathcal{Q}_1$	−0.097* (−1.865)	0.008 (0.314)
$\mathcal{Q}_2$	−0.116** (−2.409)	0.021 (0.711)
ϑ_{BMI}	0.353** (2.368)	0.402** (2.513)
ϑ_{GLD}	−0.165* (−1.932)	−0.213** (−2.432)
ϑ_{ENR}	−0.084* (−1.709)	−0.173* (−1.922)
ϑ_{VIX}	−0.386** (−2.431)	−0.342*** (−4.465)
ϑ_{USX}	0.176 (1.087)	0.297 (0.894)
P-value of Hansen's J-test	0.139	0.158
Adjusted R^2	0.381	0.342

Notes: Columns 1 and 2 of this table report estimation results of the Bitcoin returns-realized skewness and Bitcoin returns-realized kurtosis relations, respectively. α is an intercept term. $\mathcal{Q}_1$ and $\mathcal{Q}_2$ are the coefficient estimates related to the contemporaneous and lagged values of a specific higher-order moment (skewness or kurtosis), respectively. In each regression, the autoregressive lag length is determined using the AIC criterion. The values in parentheses are heteroskedasticity- and autocorrelation-consistent (HAC) t -statistics calculated using the Newey and West (1987) standard errors. The asterisks ***, **, and * denote statistical significance at the 0.01, 0.05, and 0.10 levels, respectively.

7. Conclusion

Although there is a plethora of research investigating the presence and significance of a risk-return relation for several investment assets, this topic remains unexplored in Bitcoin markets. Attempting to fill this void in the literature, the current study aims to provide empirical evidence on the relation between Bitcoin returns and volatility, within both contemporaneous and intertemporal frameworks, using high-frequency price data. To assess the sensitivity of the risk-return relation to different volatility measures, the intraday price variability of Bitcoin is proxied by four alternative measures, including realized variance, jump variation, downside realized semivariance, and negative signed jump variation. These proxy measures can successfully capture the full information embedded in Bitcoin price movements. The empirical investigation is carried out in the context of the GMM estimation, after controlling for an array of important factors common to the world financial system.

With respect to the contemporaneous risk-return relation, the results indicate that realized variance, jump variation, downside realized semivariance, and negative signed jump variation are negatively associated with Bitcoin price changes. In terms of magnitude, downside realized semivariance (negative signed jump variation) seems to have the largest (smallest) negative effect on Bitcoin returns. These findings remain qualitatively the same after including control variables in the regression analysis. On the other hand, there is a negative intertemporal relation between Bitcoin returns and realized variance, jump variation, and downside realized semivariance, but the evidence is statistically weak. The negative signed jump variation is found to have no significant intertemporal effect on returns. The addition of control variables improves the statistical significance of the coefficient estimate on downside realized semivariance. For the other volatility coefficients, the statistical significance is essentially unchanged.

All in all, this empirical evidence contributes to a better understanding of the realized volatility dynamics of Bitcoin and provides practical implications for traders as well as international portfolio managers contemplating the viability of Bitcoin as an alternative investment class. In particular, the results underscore the distinct role of jump variation and downside realized semivariance measures in explaining the risk-return relation of Bitcoin. Therefore, these realized measures should be incorporated when modeling return dynamics, since they appear to have useful information for risk management and portfolio construction decisions. Given the sensitivity of Bitcoin exchange rates to shifts in both the global stock market index and the forward-looking index of market risk, investors must keep a vigilant eye on either index in search of clues about the future path of Bitcoin prices. Moreover, the irrelevance of either gold or energy to Bitcoin price movements implies that the virtual currency may possibly act as a hedge asset against price swings in gold and energy markets.

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Erratum regarding missing Declaration of Competing Interest statements in previously published articles

Declaration of Competing Interest statements were not included in the published version of the following articles that appeared in previous issues of Journal of Economics and Business.

The appropriate Declaration/Competing Interest statements, provided by the Authors, are included below.

1. "Credit portability and spreads: Evidence in the Brazilian market" [Journal of Economics and Business, 2019; 106: 105844] <https://doi.org/10.1016/j.jeconbus.2019.05.001>

Declaration of competing interest: The authors were contacted after publication to request a Declaration of Interest statement.

2. "Does proximity to the frontier facilitate FDI-spawned spillovers on innovation and productivity?" [Journal of Economics and Business, 2018; 97: 39-49] <https://doi.org/10.1016/j.jeconbus.2018.03.002>

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3. "Learning expectations using multi-period forecasts" [Journal of Economics and Business, 2018; 102: 1-25] <https://doi.org/10.1016/j.jeconbus.2018.09.002>

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4. "The impact of uncertainty on the number of businesses" [Journal of Economics and Business, 2019; 105: 105840] <https://doi.org/10.1016/j.jeconbus.2019.04.001>

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5. "Financial development, remittances, and poverty reduction: Empirical evidence from a macroeconomic viewpoint" [Journal of Economics and Business, 2018; 96: 59-68] <https://doi.org/10.1016/j.jeconbus.2017.12.001>

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6. "Bank size, returns to scale, and cost efficiency" [Journal of Economics and Business, 2019; 105: 105842] <https://doi.org/10.1016/j.jeconbus.2019.04.003>

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7. "Interest on reserves and Federal Reserve unwinding" [Journal of Economics and Business, 2018; 97: 28-38] <https://doi.org/10.1016/j.jeconbus.2018.02.002>

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12. "Is there a risk-return trade-off in cryptocurrency markets? The case of Bitcoin" [Journal of Economics and Business, 2019; 108: 105886] <https://doi.org/10.1016/j.jeconbus.2019.105886>

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